(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
17 December 2020 (17.12.2020)

WIPO | PCT

(10) International Publication Number
WO 2020/250245 A1

(51) International Patent Classification:
A61K 36/9066 (2006.01) *A61K 31/00* (2006.01)
A61K 47/00 (2006.01)

(21) International Application Number:
PCT/IN2020/050519

(22) International Filing Date:
11 June 2020 (11.06.2020)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:
201941023055 11 June 2019 (11.06.2019) IN

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(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Declarations under Rule 4.17:
— *as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))*
— *as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))*
— *of inventorship (Rule 4.17(iv))*

Published:
— *with international search report (Art. 21(3))*
— *in black and white; the international application as filed contained color or greyscale and is available for download from PATENTSCOPE*

(54) Title: BIOAVAILABLE TURMERIC COMPOSITION AND PROCESS FOR PREPARATION THEREOF

(57) Abstract: The invention disclosed herein is a turmeric composition having enhanced self-dispersibility and bioavailability comprising a fresh turmeric rhizome extract and dried turmeric rhizome extract (standardized to 35-95% total curcuminoids), and is devoid of any externally added excipients/bio-enhancing agents/dispersing agents. The invention also disclosed herein a process for preparation of said bioavailable turmeric composition.

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"BIOAVAILABLE TURMERIC COMPOSITION AND PROCESS FOR PREPARATION THEREOF"

Technical field of the Invention:

The present invention relates to a turmeric composition, free of excipients and external bio-enhancing agents, having enhanced self-dispersibility and bioavailability. The present invention also relates to a process for preparation of said turmeric composition.

Background and Prior art of the Invention:

Natural plant products have been used throughout human history for various purposes. Currently the use of natural products in food and medicine is also increased. This has led to an intense research in to plant extracts and their health benefits. Traditionally known benefits of plants are being analysed and validated for pharmacological benefits. At the same time they are compared against modern pharmaceutical drugs for their efficacy. Many of the natural products have pharmacological or biological activity that can be exploited in pharmaceutical drug discovery and drug design. Medicines derived from plants have played an essential role in the health care of many cultures, both ancient and modern.

One of the well-known plants, extensively used in food and nutraceutical industry is turmeric. Turmeric (*Curcuma longa*) is a member of ginger family (*Zingiberaceae*). It is a perennial rhizomatous herbaceous plant native to southern Asia, extensively cultivated in all parts of India. Turmeric is extensively used in south Asian cuisine, particularly in India. Turmeric has long history of use for its medicinal properties in traditional remedies, particularly in Ayurveda such as anti-inflammatory, anti-oxidant, anti-microbial, anti-arthritic and anti-cancer activities.

Curcuminoids are the most active components of turmeric. They are the primary non-volatile components of turmeric. Curcuminoids are polyphenolic pigments and include curcumin, demethoxycurcumin, and bisdemethoxycurcumin.

Curcumin is a hydrophobic polyphenol derived from the rhizome of the herb *Curcuma longa* has a wide spectrum of biological and pharmacological activities. Curcumin which make up for about 80% of the curcuminoids, is the most studied component of turmeric which is also responsible for giving turmeric its unique yellow colour. Curcumin is known for its anti-inflammatory and antioxidant activities; curcumin has also shown activities to reduce oxidative damage related to aging.

Curcuminoids 95% is the most common form of curcuminoids extract composition available in the market. This is a concentrate of 95% curcuminoids, however, the bioavailability of these compositions are very poor.

The established pharmacological safety and efficacy of curcumin makes it a potential compound for treatment and prevention of a wide variety of human diseases, however, curcumin has not yet been considered as a therapeutic agent and the bioavailability of curcumin has been highlighted as a major problem for this.

There are different causes for reduced bioavailability of any agent within the body such as low intrinsic activity, poor absorption, high rate of metabolism, inactivity of metabolic products and/or rapid elimination and clearance from the body. However, the research carried out over past few years on turmeric revealed that poor absorption, rapid metabolism and elimination of curcumin are major reasons that severely decrease its bioavailability.

To improve the bioavailability of curcumin, numerous approaches have been undertaken.

US6440468 disclose a process for preparing a composition for oral administration, the composition containing an apolar extract and a polar extract from *Curcuma longa* rhizome. The process for obtaining the apolar extract

comprises: (a) extracting the rhizomes with an organic solvent; (b) filtration and evaporation to dryness of the extract; (c) dissolution of the oleoresin obtained in hot conditions, precipitation while allowing to cool down and filtration of the solid; (d) drying and recrystallizing the solid in order to obtain a product having a purity in curcuminoids higher than 90%. The polar extract comprises: (a) extraction of the rhizomes with water at 50-70°C and (b) filtration and evaporation of the water. However, US'468 use the solvents for preparation of apolar and polar extract and the percentage yield of obtained curcuminoids using said methods is less (around 8%).

US2014010903, its corresponding 2686/CHE/2012 and research article of Krishnakumar, Dinesh Kumar, Eapen Ninan, et al., titled, 'Enhanced absorption and pharmacokinetics of fresh turmeric (*Curcuma Longa L*) derived curcuminoids in comparison with the standard curcumin from dried rhizomes' published in Journal of Functional Foods. 2015, Vol.17, p.55 relates to a curcuminoid composition with enhanced bioavailability derived from fresh turmeric rhizomes and a process for its preparation. However, the process for preparation of composition disclosed and claimed in US'903 is essentially comprises of ultrasonication and enzyme treatment. The enzyme treatment enhances the juice yield, clarity of juice and curcuminoid concentration in the juice. Further ultrasonication after enzyme treatment is effective for enhancing the curcuminoid concentration in the resulting juice and after processing, the percentage yield of obtained juice powder is around 6 to 10% and percentages of curcumin content in juice powder is around 2 to 7%.

WO2015/025263 discloses a curcumin composition for increasing the bioavailability of curcumin, which consists of curcumin mixture and water extract in a ratio of 70:30. The curcumin mixture comprises curcumin dry crystals, volatile oil, fixed oil whereas water extract comprises soluble proteins, dietary fibers and carbohydrates extracted from turmeric. The composition also consists of a natural emulsifier isolated from *Quillaja saponaria* and lecithin.

The prior art process or methods for extraction of curcuminoids uses different organic solvents, lengthy and costly process steps. Moreover, the product thus obtained will have low yield, less curcuminoids content and poor bioavailability. The water extract of fresh rhizome will lead to the product with very low levels of curcuminoids. Also, the compositions in the art are generally comprises volatile oil, fixed oil, emulsifiers and external bio-enhancing agents such as PVP and polysorbate which may have safety issues. There is no disclosure of any bioavailable turmeric composition which is devoid of added excipients or solvent or volatile/fixed oil or gum in the art.

Thus, there is still a need exist in the art to provide a turmeric composition comprising higher percentages of curcuminoids, free of oils, any excipients and external bio-enhancing agents, however, achieves improved solubility/dispersibility, thereby providing enhanced bioavailability.

Also in general, if a drug is not soluble in water or has a very low solubility it cannot be absorbed through the cell membranes and thus its absorption will be negligible and consequently its bioavailability will be too low.

Object of the Invention:

In view of the above, the inventors of the present invention have come up with a bioavailable Turmeric composition comprising turmeric extract containing higher amounts of curcuminoids prepared by environmental friendly process of physical methods and co-boiling/homogenization of fresh turmeric rhizome extract and dried turmeric rhizome extract (standardized to 35-95% total curcuminoids) at higher temperature, without use of any excipients and external bio-enhancing agents, still achieving enhanced self-dispersibility and bioavailability. The final composition comprises 100% turmeric derived composition.

Summary of the Invention:

In accordance with the above objective, the present invention discloses a 100% pure turmeric composition having higher percentage of curcuminoids with enhanced self-dispersibility and bioavailability.

In a preferred aspect, the present invention discloses a highly bioavailable and self-dispersible turmeric composition comprising a fresh turmeric rhizome extract and dried turmeric rhizome extract standardized to 35-95% total curcuminoids, in a ratio ranges from 0.5:50 to 50:0.5 v/w.

In another aspect, the present invention discloses a turmeric composition that is free of any externally added excipients, bio-enhancing agents, emulsifiers, dispersing agents, solvents, fixed oils, volatile oils or gums.

In another preferred aspect, the present invention discloses a turmeric composition comprising curcuminoids that contains Curcumin, Demethoxycurcumin (DMC) and Bisdemethoxycurcumin (BDMC) in their native profile.

In yet another preferred aspect, the present invention discloses a process for preparation of said turmeric composition comprising homogenizing/wet milling/colloidal milling of fresh turmeric rhizome extract and dried turmeric rhizome extract standardized to 35-95% total curcuminoids together at higher temperature followed by drying and powdering to get the free-flowing powder of self-dispersible turmeric composition.

In yet another aspect, the present turmeric composition enhances the bioavailability of Curcumin, Demethoxycurcumin (DMC), Bismethoxycurcumin (BDMC) and Tetrahydrocurcumin (THC), an active metabolite of Curcumin.

In yet another aspect, the present turmeric composition enhances the bioavailability of DMC, BDMC and THC significantly higher than unformulated standard curcuminoids (C-95) extract.

Brief description of Figure(s):

Figure 1: Release profile of Turmeric Composition 1 of **Example 1**

Figure 2: Dispersion study of Turmeric Composition 1

Figure 3: Mean plasma Total Curcumin Concentration Vs. Time graph of Turmeric Composition 1

Figure 4: Mean plasma Demethoxycurcumin (DMC) Concentration Vs. Time graph of Turmeric Composition 1

Figure 5: Mean plasma Bismethoxycurcumin (BDMC) Concentration Vs. Time graph of Turmeric Composition 1

Figure 6: Mean plasma Tetrahydrocurcumin (THC) Concentration Vs. Time graph of Turmeric Composition 1

Figure 7: Release profile of Turmeric Composition 2 of **Example 8**

Figure 8: Release profile of Turmeric Composition 3 of **Example 12**

Figure 9: Release profile of Turmeric Composition 4 of **Example 16**

Figure 10: Release profile of Turmeric Composition 5 of **Example 20**

Figure 11: Release profile of Turmeric Composition 1 of **Example 1** *vis-à-vis* standard curcumin extract (95% total curcuminoids) and fresh turmeric rhizome extract

Figure 12: Release profile of Turmeric Composition 1 of **Example 1** Vs. Marketed Product

Detailed description of the Invention:

The invention will now be described in detail in connection with certain preferred and optional embodiments, so that various aspects thereof may be more fully understood and appreciated.

Source and Geographical origin of the biological material used in the Invention:

Curcuma longa (Turmeric)
Geographical Origin: Native to the Indian subcontinent and Southeast Asia
Source: The turmeric rhizome was procured from local vendor (farmer) of Erode District, Tamil Nadu, India.

The term 'Standardized turmeric rhizome extract' refers herein is the dry turmeric rhizome extract which is standardized to 30-95% total curcuminoids and includes partially purified Oleoresin extract and extract with 95% Total Curcuminoids.

The term 'Self-dispersible' refers herein to spontaneous dispersion of a product into aqueous phase forming uniform suspension without any external force applied to make it disperse.

The present invention discloses a 100% pure turmeric composition having higher percentage of curcuminoids with enhanced self-dispersibility and bioavailability.

In a preferred embodiment, the present invention discloses a highly bioavailable and self-dispersible turmeric composition comprising a fresh turmeric rhizome extract and dried turmeric rhizome extract (standardized to 35-95% total curcuminoids), in a ratio ranges from 0.5:50 to 50:0.5 v/w; wherein said composition is devoid of any externally added excipients, bio-enhancing agents, emulsifiers, dispersing agents, solvents, fixed oils, volatile oils or gums.

In another embodiment, the turmeric composition of the present invention comprises 20 to 85% of total curcuminoids content in self-dispersible form, 0.25 to <4% of proteins content and 0.25 to <49% of carbohydrates content.

In another embodiment, the present invention discloses a method for enhancing the bioavailability of curcuminoids using solvent free fresh turmeric rhizome extract; wherein said method comprising extraction of solvent free fresh turmeric rhizome extract, homogenizing/wet milling/colloidal milling of fresh turmeric

rhizome extract and dried turmeric rhizome extract standardized to 35-95% total curcuminoids together at higher temperature followed by drying and powdering to get the free-flowing powder of self-dispersible turmeric composition.

In a preferred embodiment, the present invention discloses a cost effective, eco-friendly process for preparation of turmeric composition comprising the steps of;

a) slicing the fresh turmeric rhizome and washing with water followed by squeezing to separate the liquid extract and filtering the extract;
b) heating the extract of step (a) upto 60^0C with continuous stirring for 30 minutes;
c) adding dried turmeric rhizome extract (standardized to 35-95% total curcuminoids) to the extract of step (b) in the ratio ranging from 0.5:50 to 50:0.5;
d) continuously homogenizing the mixture of step (c) for 30 to 60 minutes between 25°C to 60°C at 1400 RPM to obtain slurry; and
e) vacuum drying the slurry of step (d) at 60°C to 85°C for 8 hours followed by milling the flakes to obtain free flowing powder of self-dispersible turmeric composition.

Accordingly, the process for preparation of present turmeric composition comprising the steps of;

a) slicing the fresh turmeric rhizome and washing with RO water followed by squeezing using low screw extruder with 0.5mm to 1mm mesh to separate the liquid extract and filtering the extract;
b) heating the extract obtained in step (a) upto 60^0C with continuous stirring for 30 minutes using a homoziniser;
c) slowly adding dried turmeric rhizome extract standardized to 35-95% total curcuminoids to the extract of step (b) in the ratio ranging from 0.5:50 to 50:0.5; with continuous homogenizing for 30 to 60 minutes at 1400 RPM between 25°C to 60°C to obtain viscous slurry;

d) vacuum drying the slurry of step (c) at 60°C to 85°C, 650 mmHg vacuum for 8 hours by Rotary Vacuum Dryer (RVD); and
e) milling the flakes of step (d) to obtain free flowing powder (particle size >120 mesh) of self-dispersible turmeric composition.

In the present process fresh turmeric rhizome was washed with the RO water and squeezed using low speed screw extruder with 0.5mm to 1 mm mesh to separate the liquid extract and filtered the extract. The extract obtained was heated upto 60°C with continuous stirring for 30 minutes using a homoziniser. Dried turmeric rhizome extract standardized to 35-95% total curcuminoids was slowly added to the fresh turmeric rhizome extract and continuously homogenized for 30 to 60 minutes at 1400 RPM between 25°C to 60°C to obtain viscous slurry. The slurry was vacuum dried at 60°C to 85°C, 650 mmHg vacuum for 8 hours by Rotary Vacuum Dryer (RVD) and milled using comminuting mill to get free flowing powder (particle size >120 mesh). This final free flowing powder contains >50% total curcuminoids and is highly self-dispersible and bioavailable.

The percentage yield of fresh extract from fresh turmeric rhizome is ranging from 50 to 80% with a solid content ranging from 5 to 13%.

In another embodiment, the fresh turmeric rhizome extract obtained by using low speed screw extruder with 0.5mm to 1mm mesh is a solvent free extract comprising <4% hydrophobic contents and acts as dispersing agent.

In another embodiment, the fresh turmeric rhizome extract can be dried into powder for storage which can be reconstituted with water and used.

The ratio of fresh turmeric rhizome extract to dried turmeric rhizome extract standardized to 35-95% total curcuminoids is 0.5:50 to 50:0.5 v/w.

In another embodiment, the dried turmeric rhizome extract standardized to 35-95% total curcuminoids is extracted from dried turmeric rhizome using solvents such as ethanol, ethyl acetate, methanol, acetone and hexane.

The homogenization or wet milling or colloidal milling of fresh turmeric rhizome extract and dried turmeric rhizome extract standardized to 35-95% total curcuminoids at higher temperature enhances the self-dispersion and bioavailability of curcuminoids.

The bioavailable turmeric composition of present invention comprises higher amount of curcuminoids (20 to 85%) and are amorphous in nature.

In another preferred embodiment, the present turmeric composition comprises curcuminoids that contains curcumin in an amount ranges from 15 to 74%; Demethoxycurcumin (DMC) in an amount ranges from 10 to 30%, and Bisdemethoxycurcumin (BDMC) in an amount ranges from 3 to 20%.

In another embodiment, the present turmeric composition enhances the bioavailability of Curcumin, Demethoxycurcumin (DMC), Bismethoxycurcumin (BDMC) and Tetrahydrocurcumin (THC), an active metabolite of Curcumin.

In another embodiment, the present turmeric composition enhances the bioavailability of DMC, BDMC and THC significantly higher than unformulated standard curcuminoids (C-95) extract.

In another embodiment, the present turmeric composition delivers higher levels of DMC, BDMC and THC into the blood compared to unformulated standard curcuminoids.

In another embodiment, the present invention discloses the release profile of turmeric composition; wherein the results exhibit the increased solubility and

sustained release for present composition compared to unformulated standard curcuminoids (having 95% total curcuminoids, C-95). The same is depicted in **Figure 1** and **Figures 7 to 10**.

In another embodiment, the present invention discloses the comparative self-dispersion study between the present turmeric composition and unformulated standard curcuminoids (having 95% total curcuminoids, C-95); wherein the result show that present composition exhibits enhanced self-dispersion compared to the unformulated standard curcuminoids which floats on top of the water. The dispersion result is given in **Figure 2**.

In another embodiment, the present invention discloses the bioavailability and pharmacokinetics profile of present turmeric composition; wherein the results exhibit that the present turmeric composition was found to be superior to unformulated standard curcuminoids (standardized to 95% total curcuminoids) in enhancing the bioavailability of total Curcumin, Demethoxycurcumin (DMC), Bisdemethoxycurcumin (BDMC) and Tetrahydrocurcumin (THC). The same is depicted in **Figure 3** to **Figure 6**.

In another embodiment, the present invention discloses synergy of the present turmeric composition; wherein the results show that present composition exhibits increased release profile compared to 'unformulated standard curcuminoids (having 95% total curcuminoids, C-95)' and 'fresh turmeric rhizome extract powder'. The same is depicted in **Figure 11.**

In another embodiment, the present invention discloses efficacy of the present turmeric composition; wherein the result shows that present composition exhibits increased release profile compared to 'marketed product' (composition comprising fresh rhizome juice powder as disclosed in Patent No. US2014010903). The same is depicted in **Figure 12.**

In another embodiment, the turmeric composition of the present invention is devoid of any externally added excipients, dispersing agents, bio-enhancing agents, emulsifiers, solvents, fixed oils, volatile oils or gums and having enhanced self-dispersibility and bioavailability.

The present bioavailable turmeric composition is in the form of powder which can be formulated into various dosage forms such as tablets, capsules, pills, solutions, pastes, lozenges and the like.

In another embodiment, the present invention discloses the use of bioavailable turmeric composition in the treatment of inflammatory diseases, cognitive diseases, eye diseases, skin diseases, stress and the like.

In another embodiment, the present invention discloses a method of treating inflammatory diseases, cognitive diseases, eye diseases, skin diseases and stress by administering therapeutically effective amount of present bioavailable turmeric composition to a subject in need thereof; wherein the therapeutically effective amount is 500mg/per day.

The recommended dosage of present bioavailable composition given to a subject in need is 500 mg; wherein the subject is human or animal.

In another optional embodiment, the present bioavailable turmeric composition can also be prepared by using fresh extract from rhizomes such as ginger rhizome, fresh extract from vegetables such as potato, beet root and fresh extract from fruit such as guava instead of fresh turmeric rhizome extract.

Example:

Some typical examples illustrating the embodiments of the present invention are provided; however, these examples are exemplary only and should not be regarded as limiting the elements of the present invention.

Example 1: Turmeric Composition 1

Sr. No.	Ingredients	Quantity (g)
1.	Turmeric rhizome extract powder (Standardized)	90
2.	Fresh turmeric rhizome extract (on dry basis)	10
	Total	100

Example 2: Components in Turmeric Composition 1

Sr. No.	Components	Quantity (%)
1.	Curcuminoids content	50 to 94%
2.	Proteins content	0.25 to <4%
3.	Carbohydrates content	0.25 to <49%

Example 3: Process for preparation of Turmeric Composition 1

a) Washing 2.82 kg of fresh turmeric rhizome with RO water followed by squeezing using low speed screw extruder with 0.5mm to 1 mm mesh to separate the liquid extract and filtering the extract;

b) heating the extract obtained in step (a) upto 60°C with continuous stirring for 30 minutes using a homoziniser;

c) adding 1.8 kg of dried turmeric rhizome extract (standardized to 55% total curcuminoids) slowly to the extract of step (b) with continuous homogenizing for 30 to 60 minutes at 1400 RPM between 25°C to 60°C to obtain viscous slurry;

d) vacuum drying the slurry of step (c) at 60°C to 85°C, 650 mmHg vacuum for 8 hours using Rotary Vacuum Dryer (RVD); and

e) milling the flakes of step (d) to obtain free flowing powder (particle size >120 mesh) of self-dispersible turmeric composition.

Yield of fresh extract: 1.8kg

Example 4: Release profile of Turmeric Composition 1

The composition was tested for solubility of curcuminoids and its release in buffer. 500mg of Composition 1 of **Example 1** (Test product) was added into the 400 ml phosphate buffer (pH 6.8) under stirring at 37°C. Samples were drawn for up to four hours, filtered/centrifuged and tested for curcumin content by measuring absorbance at 424nm using UV-Vis spectrophotometer. The result shows the increased solubility and release profile of Composition 1 compared to unformulated standard curcumin extract [standardized to 95% Curcuminoids (C-95)] (Reference product). The absorbance vs. time graph was plotted as illustrated in **Figure 1**.

Example 5: Curcuminoids content of Turmeric Composition 1

The composition was tested for total Curcuminoids and the result is given below:

Total Curcuminoids Assay by HPLC	
Curcumin	30.46%
Desmethoxycurcumin	12.54%
Bisdemethoxycurcumin	9.70%
Total Curcuminoids	52.70%

Example 6: Comparative self-dispersion profile of Turmeric Composition 1 *vis-à-vis* Standard Curcuminoids (95% total curcuminoids)

A comparative self-dispersion study was carried out between the Composition 1 (Test product) and unformulated Standard Curcuminoids having 95% total curcuminoids (Reference product, C-95). The dispersion is given in **Figure 2**. The result shows that the test product showed enhanced self-dispersion compared to the reference product. There was no self-dispersion/dispersion observed for reference product

Example 7: Comparative Bioavailability studies of the Turmeric Composition 1 *vis-à-vis* Standard Curcuminoids (95% total curcuminoids)

A comparative oral bioavailability study of Composition 1 (Test product) and unformulated Standard Curcuminoids (standardized to 95%) (Reference product) was carried out in healthy adult human males (n=18). Relative bioavailability and pharmacokinetics of total curcumin was evaluated following oral administration of single dose of test product equivalent to 2 g Curcuminoids Vs. unformulated standard Curcuminoids (standardized to 95%) equivalent to 2 g Curcuminoids. The blood samples were collected at 00.00, 00.25, 00.50, 01.00, 01.50, 02.00, 02.50, 03.00, 03.50, 04.00, 06.00, 08.00, 12.00, 16.00 and 24.00 hours post dose and were centrifuged at 3800 rpm for 10 minutes at 2°C to 8°C for separating the plasma. The plasma samples were treated with β-Glucuronidase/sulfatase (EC 3.2.1.31) from Helix Pomatia in 0.1M phosphate for 1 hour at 37°C to ensure the complete hydrolysis of glucuronide/sulfate conjugates of Curcuminoids. Then, the curcuminoids were extracted with 2 ml of extraction solvent [Ethyl Acetate: Acetonitrile (95:05)] and evaporated to dryness under the stream of nitrogen. The dried residue was reconstituted in 0.5 ml of mobile phase and analyzed for total curcumin demethoxycurcumin (DMC) and bisdemethoxycurcumin (BDMC) and tetrahydrocurcumin (THC) using LC-MS/MS.

Total Curcuminoids (Curcumin, Demethoxycurcumin and Bisdemethoxycurcumin) were quantified against respective USP standard. Curcumin-D6 was used as internal standard (ISTD). The mean total plasma Curcumin, Demethoxycurcumin, Bisdemethoxycurcumin and Tetrahydrocurcumin concentration vs. time graphs for 'Test product' and 'Reference product' was plotted as illustrated in **Figures 3, 4, 5 and 6** respectively.

Pharmacokinetic Results: The pharmacokinetic result of Turmeric Composition 1 is given in below **Table 1 to Table 4** as follows:

Table 1: Mean (SD) of Pharmacokinetic Parameters of Total Curcumin

PK Parameter (Units)	**Test product**	**Reference Product**
C_{max} (ng/mL)	193.22 (51.86)	7.62 (2.64)
AUC_{0-t} (ng.hr/mL)	1225.82 (694.37)	24.77 (18.21)
T_{max} (hr)*	4.0 (3.50, 16.00)	2.00 (0.50, 12.00)

*For T_{max}, Median (Min, Max) are presented

Table 2: Mean (SD) of Pharmacokinetic Parameters of Demethoxycurcumin (DMC)

PK Parameter (Units)	**Test Product**	**Reference Product**
C_{max} (ng/mL)	125.20 (34.71)	7.12 (3.73)
AUC_{0-t} (ng.hr/mL)	856.94 (570.43)	19.68 (13.78)
T_{max} (hr)*	4.00 (3.50, 16.00)	1.50 (0.50, 3.50)

*For T_{max}, Median (Min, Max) are presented

Table 3: Mean (SD) of Pharmacokinetic Parameters of Bisdemethoxycurcumin (BDMC)

PK Parameter (Units)	**Test Product**	**Reference Product**
C_{max} (ng/mL)	32.11 (16.64)	3.81 (1.70)
AUC_{0-t} (ng.hr/mL)	256.95 (206.73)	5.49 (3.00)
T_{max} (hr)*	3.00 (1.50, 8.00)	1.50 (0.50, 3.50)

*For T_{max}, Median (Min, Max) are presented

Table 4: Mean (SD) of Pharmacokinetic Parameters of Tetrahydrocurcumin (THC)

PK Parameter (Units)	**Test Product**	**Reference Product**
C_{max} (ng/mL)	10.95 (4.59)	0.66 (0.30)
AUC_{0-t} (ng.hr/mL)	44.17 (24.23)	1.427 (0.83)
T_{max} (hr)*	3.00 (1.50, 6.00)	1.50 (1.00, 3.00)

*For T_{max}, Median (Min, Max) are presented

C_{max}: Maximum measured Plasma concentration

AUC_{0-t}: The area under the Plasma concentration versus time curve from time '0' to specified time

T_{max}: Time of the maximum measured plasma concentration (Minimum and Maximum median is presented).

The mean plasma Total Curcumin concentration Vs. time graph for 'Test product' and 'Reference product' is given in **Figure 3**. The pharmacokinetic result is given in **Table 1**. The results from **Table 1** showed that the C_{max} and AUC_0-t of the test product for Total Curcumin was 25.35 times and 49.48 times higher than the C_{max} and AUC_{0-t} of the Reference product (unformulated Standard Curcuminoids) respectively.

The mean plasma Demethoxycurcumin (DMC) concentration Vs. time graph for 'Test product' and 'Reference product' is given in **Figure 4**. The pharmacokinetic result is given in **Table 2**. The results from **Table 2** showed that the C_{max} and AUC_{0-t} of the Test product for Demethoxycurcumin was 17.58 times and 43.53 times higher than the C_{max} and AUC_{0-t} of the Reference product (unformulated Standard Curcuminoids) respectively.

The mean plasma Bisdemethoxycurcumin (BDMC) concentration Vs. time graph for 'Test product' and 'Reference product' is given in **Figure 5**. The

pharmacokinetic result is given in **Table 3**. The results from **Table 3** showed that the C_{max} and AUC_{0-t} of the test product for Bisdemethoxycurcumin was 8.41 times and 46.79 times higher than the C_{max} and AUC_{0-t} of the Reference product (unformulated Standard Curcuminoids) respectively.

The mean plasma Tetrahydrocurcumin (THC) concentration Vs. time for 'Test product' and 'Reference product' is given in **Figure 6**. The pharmacokinetic result is given in **Table 4**. The results from **Table 4** showed that the C_{max} and AUC_0-t of the test product for Tetrahydrocurcumin was 16.71 times and 30.95 times higher than the C_{max} and AUC_{0-t} of the Reference product (unformulated Standard Curcuminoids) respectively.

It was concluded from above **Figures 3 to 6** and **Tables 1 to 4** that, the 'Test product' was found to superior to 'Reference product' for enhancing the bioavailability of total curcumin, Demethoxycurcumin (DMC), Bisdemethoxycurcumin (BDMC) and Tetrahydrocurcumin (THC). Plasma concentrations of DMC and BDMC were surprisingly found to be significantly higher.

Example 8: Turmeric Composition 2

Sr. No.	Ingredients	Quantity (g)
1.	Turmeric rhizome extract powder (Standardized)	90
2.	Fresh turmeric rhizome extract (on dry basis)	10
	Total	100

Example 9: Process for preparation of Turmeric Composition 2

a) Washing 150gm of fresh turmeric rhizome with RO water followed by squeezing using low speed screw extruder with 0.5mm to 1 mm mesh to separate liquid extract and filtering the extract;
b) heating the extract obtained in step (a) upto 60°C with continuous stirring for 30 minutes using a homoziniser;

c) adding 90g of dried turmeric rhizome extract (standardized to 60% total curcuminoids) slowly to the extract of step (b) with continuous homogenizing for 30 to 60 minutes at 1400 RPM between 25°C to 60°C to obtain viscous slurry;
d) vacuum drying the slurry of step (c) at 60°C 85°C, 650 mmHg vacuum for 8 hours by Rotary Vacuum Dryer (RVD); and
e) milling the flakes of step (d) to obtain free flowing powder (particle size >120 mesh) of self-dispersible turmeric composition.

Yield of fresh extract: 100g

Example 10: Release profile of Turmeric Composition 2

The composition was tested for its active ingredient solubility and release in buffer. 500mg of Composition 2 of **Example 8** was added into the 400 ml phosphate buffer (pH 6.8) under intermittent stirring at 37°C. Samples were drawn for up to four hours, filtered/centrifuged and tested for curcumin content by measuring absorbance at 424nm using UV-Vis spectrophotometer. The results shows the increased solubility and release profile for Composition 2 compared to standard curcumin extract [standardized to 95% Curcuminoids (C-95%)]. The absorbance vs. time graph was plotted as illustrated in **Figure 7**.

Example 11: Curcuminoids content of Turmeric Composition 2

The composition was tested for total Curcuminoids and the result is given below:

Total Curcuminoids Assay by HPLC	
Curcumin	30.68%
Desmethoxycurcumin	12.60%
Bisdemethoxycurcumin	9.74%
Total Curcuminoids	53.02%

Example 12: Turmeric Composition 3

Sr. No.	Ingredients	Quantity (g)
1.	Turmeric rhizome extract powder (Standardized)	80
2.	Fresh turmeric rhizome extract (on dry basis)	20
	Total	100

Example 13: Process for preparation of Turmeric Composition 3

a) Slicing 250g of fresh turmeric rhizome and washing with RO water followed by squeezing using low screw extruder with 0.5 mm to 1 mm mesh to separate liquid extract and filtering the extract;
b) heating the extract obtained in step (a) upto 60°C with continuous stirring for 30 minutes using a homoziniser;
c) adding 80g of dried turmeric rhizome extract (standardized to 55% total curcuminoids) slowly to the extract of step (b) with continuous homogenizing for 30 to 60 minutes at 1400 RPM between 25°C to 60°C to obtain viscous slurry;
d) vacuum drying the slurry of step (c) at 60°C 85°C, 650 mmHg vacuum for 8 hours by Rotary Vacuum Dryer (RVD); and
e) milling the flakes of step (d) to obtain free flowing powder (particle size >120 mesh) of self-dispersible turmeric composition

Yield of fresh extract: 200g

Example 14: Release profile of Turmeric Composition 3

The composition was tested for its active ingredient solubility and release in buffer. 500mg of Composition 3 of **Example 12** was added into the 400 ml phosphate buffer (pH 6.8) under intermittent stirring at 37°C. Samples were drawn for up to five hours and tested for curcuminoid content by measuring absorbance at 424nm using UV-Vis spectrophotometer. The result shows that Composition 3 has enhanced solubility and release profile compared to unformulated standard curcumin extract [standardized to 95% Curcuminoids (C-95)]. The absorbance vs. time graph was plotted as illustrated in **Figure 8**.

Example 15: Curcuminoids content of Turmeric Composition 3

The composition was tested for total Curcuminoids and the result is given below:

Total Curcuminoids Assay by HPLC	
Curcumin	34.30%
Desmethoxycurcumin	10.43%
Bisdemethoxycurcumin	7.20%
Total Curcuminoids	51.93%

Example 16: Turmeric Composition 4

Sr. No.	**Ingredients**	**Quantity (g)**
1.	Turmeric rhizome extract powder (Standardized)	70
2.	Fresh turmeric rhizome extract (on dry basis)	30
	Total	100

Example 17: Process for preparation of Turmeric Composition 4

a) Slicing 400g of fresh turmeric rhizome and washing with RO water followed by squeezing using low screw extruder with 0.5 mm to 1 mm mesh to separate liquid extract and filtering the extract;

b) heating the extract obtained in step (a) upto 60°C with continuous stirring for 30 minutes using a homoziniser;

c) slowly adding 70g of dried turmeric rhizome extract (standardized to 95% total curcuminoids) to the extract of step (b) with continuous homogenizing for 30 to 60 minutes at 1400 RPM between 25°C to 60°C to get viscous slurry;

d) vacuum drying the slurry of step (c) at 60°C 85°C, 650 mmHg vacuum for 8 hours by Rotary Vacuum Dryer (RVD); and

e) milling the flakes of step (d) to obtain free flowing powder (particle size >120 mesh) of self-dispersible turmeric composition.

Yield of fresh extract: 300 g

Example 18: Release profile of Turmeric Composition 4

The composition was tested for its active ingredient solubility and release in buffer. 500mg of Composition 4 of **Example 16** was added into 400 ml phosphate buffer (pH 6.8) under intermittent stirring at 37°C. Samples were drawn for up to four hours and tested for curcuminoid content by measuring absorbance at 424nm using UV-Vis spectrophotometer. The result shows that Composition 4 has enhanced solubility and release profile compared to standard curcumin extract [standardized to 95% Curcuminoids (C-95)]. The absorbance vs. time graph was plotted as illustrated in **Figure 9**.

Example 19: Curcuminoids content of Turmeric Composition 4

The composition was tested for total curcuminoids and the result is given below:

Total Curcuminoids Assay by HPLC	
Curcumin	53.93 %
Demethoxycurcumin	12.17 %
Bisdemethoxycurcumin	2.26 %
Total Curcuminoids	68.36 %

Example 20: Turmeric Composition 5

Sr. No.	**Ingredients**	**Quantity (g)**
1.	Turmeric rhizome extract powder (Standardized)	85
2.	Fresh turmeric rhizome extract (on dry basis)	15
	Total	100

Example 21: Process for preparation of Turmeric Composition 5

a) Slicing 200g of fresh turmeric rhizome and washing with RO water followed by squeezing using low screw extruder with 0.5 mm to 1 mm mesh to separate liquid extract and and filtering the extract;

b) heating the extract obtained in step (a) upto 60°C with continuous stirring for 30 minutes using a homoziniser;

c) slowly adding 85g of dried turmeric rhizome extract (standardized to 95% total curcuminoids) to the extract of step (b) with continuous homogenizing for 30 to 60 minutes at 1400 RPM between 25°C to 60°C to get viscous slurry;
d) vacuum drying the slurry of step (c) at 60°C 85°C, 650 mmHg vacuum for 8 hours by Rotary Vacuum Dryer (RVD); and
e) milling the flakes of step (d) to obtain free flowing powder (particle size >120 mesh) of self-dispersible turmeric composition.

Yield of fresh extract: 150 g

Example 22: Release profile of Turmeric Composition 5

The composition was tested for its active ingredient solubility and release in buffer. 500mg of Composition 5 of **Example 20** was added into the 400 ml phosphate buffer (pH 6.8) under intermittent stirring at 37°C. Samples were drawn for up to four hours and tested for curcuminoid content by measuring absorbance at 424nm using UV-Vis spectrophotometer. The result shows that the Composition 5 has enhanced solubility and release profile compared to unformulated standard curcumin extract [standardized to 95% Curcuminoids (C-95)]. Absorbance vs. time graph was plotted as illustrated in **Figure 10**.

Example 23: Curcuminoids content of Turmeric Composition 5

The composition was tested for total Curcuminoids and the result is given below:

Total Curcuminoids Assay by HPLC	
Curcumin	62.99 %
Demethoxycurcumin	14.03 %
Bisdemethoxycurcumin	2.40 %
Total Curcuminoids	79.42%

Example 25: Release profile of Turmeric Composition 1 of Example 1 *vis-à-vis* standard curcumin extract (95% total curcuminoids) and fresh turmeric rhizome extract

The composition was tested for its active ingredient solubility and release in buffer. 500mg of Composition 1 of **Example 1** (Test product) was added into the 400 ml phosphate buffer (pH 6.8) under intermittent stirring at 37°C. Samples were drawn for up to five hours, filtered/centrifuged and tested for curcumin content by measuring absorbance at 424nm using UV-Vis spectrophotometer. The result shows the increased release profile for 'Test product' compared to 'unformulated standard curcumin extract (95% total curcuminoids)' and 'fresh turmeric rhizome extract'. Absorbance vs. time graph was plotted as illustrated in **Figure 11**.

Example 26: Release profile of Turmeric Composition 1 of Example 1 Vs. Marketed Product

The composition was tested for its active ingredient solubility and release in buffer. 500mg of Composition 1 of **Example 1** (Test product) was added into the 400 ml phosphate buffer (pH 6.8) under intermittent stirring at 37°C. Samples were drawn for up to five hours, filtered/centrifuged and tested for curcumin content by measuring absorbance at 424nm using UV-Vis spectrophotometer. The result shows the increased release profile for 'Test product' compared to the 'marketed product'. The Absorbance vs. time graph was plotted as illustrated in **Figure 12**.

We Claim,

1. A highly bioavailable and self-dispersible turmeric composition comprising a fresh turmeric rhizome extract and dried turmeric rhizome extract (standardized to 35-95% total curcuminoids), in a ratio ranges from 0.5:50 to 50:0.5 v/w;

   wherein said composition is devoid of any externally added excipients, bio-enhancing agents, emulsifiers, dispersing agents, solvents, fixed oils, volatile oils or gums.
2. The bioavailable turmeric composition as claimed in claim 1, wherein the composition comprises 20 to 85% of total curcuminoids content in self-dispersible form, 0.25 to <4% of proteins content and 0.25 to <49% of carbohydrates content.
3. The bioavailable turmeric composition as claimed in claim 2, wherein the curcuminoids content comprises curcumin in an amount of 15 to 74%; Desmethoxycurcumin (DMC) in an amount of 10 to 30%, and Bisdemethoxycurcumin (BDMC) in an amount ranges from 3 to 20%.
4. The bioavailable Turmeric composition as claimed in claims 1 to 3, wherein the composition is in the form of powder and is formulated into various dosage forms such as tablets, capsules, pills, solutions, pastes, lozenges and the like.
5. The bioavailable Turmeric composition as claimed in claims 1 to 5, wherein the composition delivers higher levels of Desmethoxycurcumin (DMC), Bisdemethoxycurcumin (BDMC) and Tetrahydrocurcumin (THC) into the blood compared to unformulated standard curcuminoids
6. A cost effective, eco-friendly process for preparation of bioavailable turmeric composition as claimed in claim 1 comprising the steps of;
   a) slicing the fresh turmeric rhizome and washing with water followed by squeezing to separate the liquid extract and and filtering the extract;

b) heating the extract of step (a) upto 60^0C with continuous stirring for 30 minutes;

c) adding dried turmeric rhizome extract (standardized to 35-95% total curcuminoids) to the extract of step (b) in the ratio ranging from 0.5:50 to 50:0.5;

d) continuously homogenizing the mixture of step (c) for 30 to 60 minutes between 25°C to 60°C at 1400 RPM to obtain slurry; and

e) vacuum drying the slurry of step (d) at 60°C to 85°C for 8 hours followed by milling the flakes to obtain free flowing powder of self-dispersible turmeric composition.

7. The process for preparation of bioavailable turmeric composition as claimed in claim 6, wherein the fresh turmeric rhizome extract obtained by using low speed screw extruder with 0.5mm to 1mm mesh is a solvent free extract comprising <4% hydrophobic contents and acts as dispersing agent.
8. The process for preparation of bioavailable turmeric composition as claimed in claim 6, wherein said composition can also be prepared by using fresh extract from rhizomes such as ginger rhizome, fresh extract from vegetables such as potato, beet root and fresh extract from fruit such as guava instead of fresh turmeric rhizome extract.
9. A method of treating inflammatory diseases, cognitive diseases, eye diseases, skin diseases and stress by administering therapeutically effective amount of bioavailable turmeric composition as claimed in claims 1 to 5 to a subject in need thereof.
10. The bioavailable turmeric composition as claimed in claims 1 to 5 useful in the treatment of inflammatory diseases, cognitive diseases, eye diseases, skin diseases and stress.

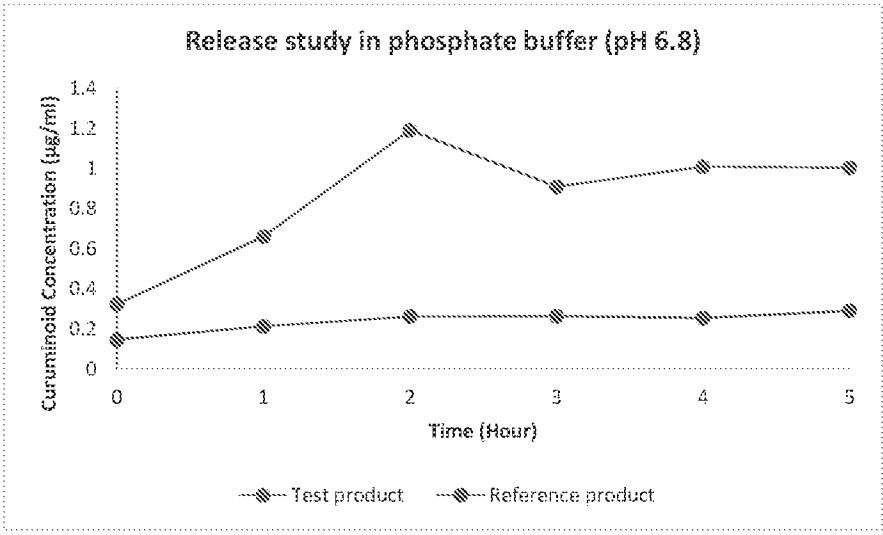
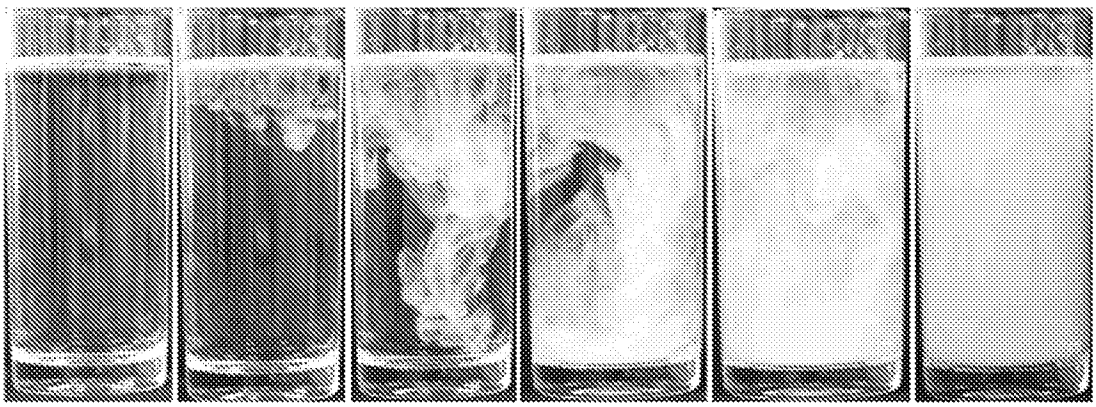
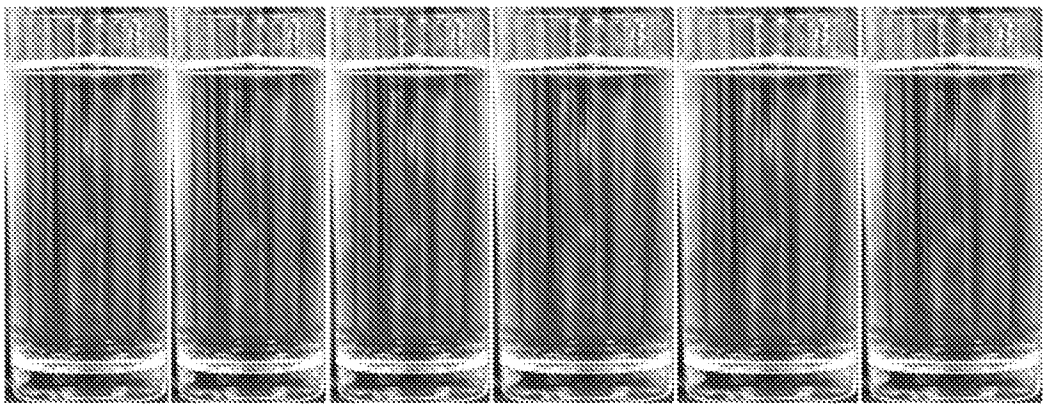


Figure 1

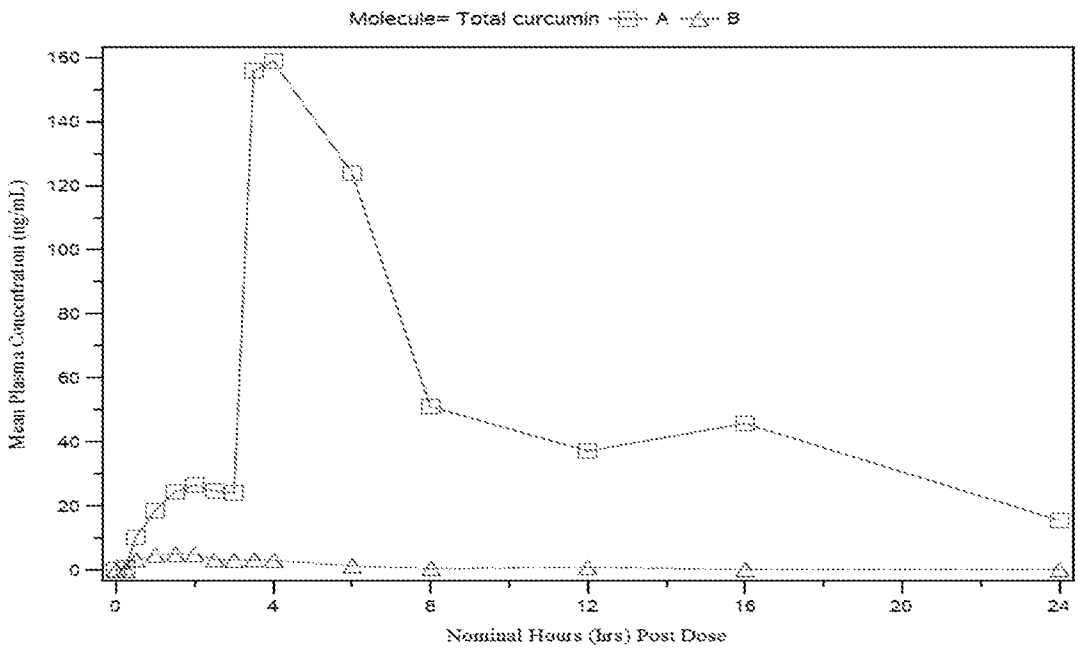


a) Test Product



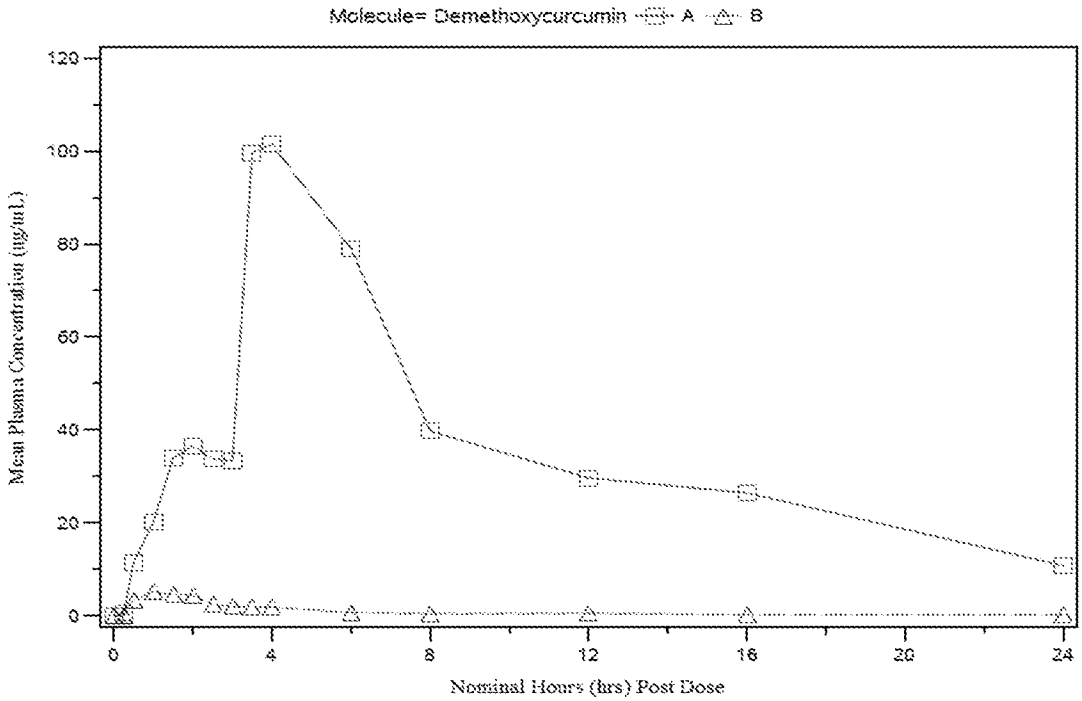
b) Reference Product

Figure 2



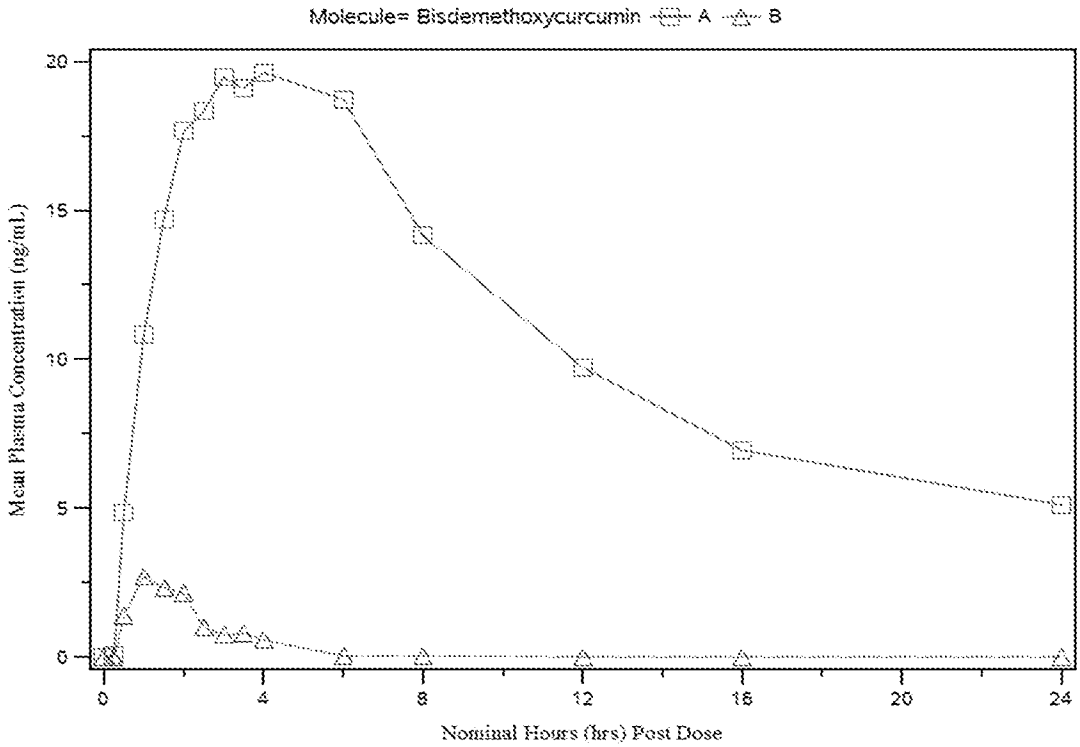
A = Test Product; B = Reference Product

Figure 3



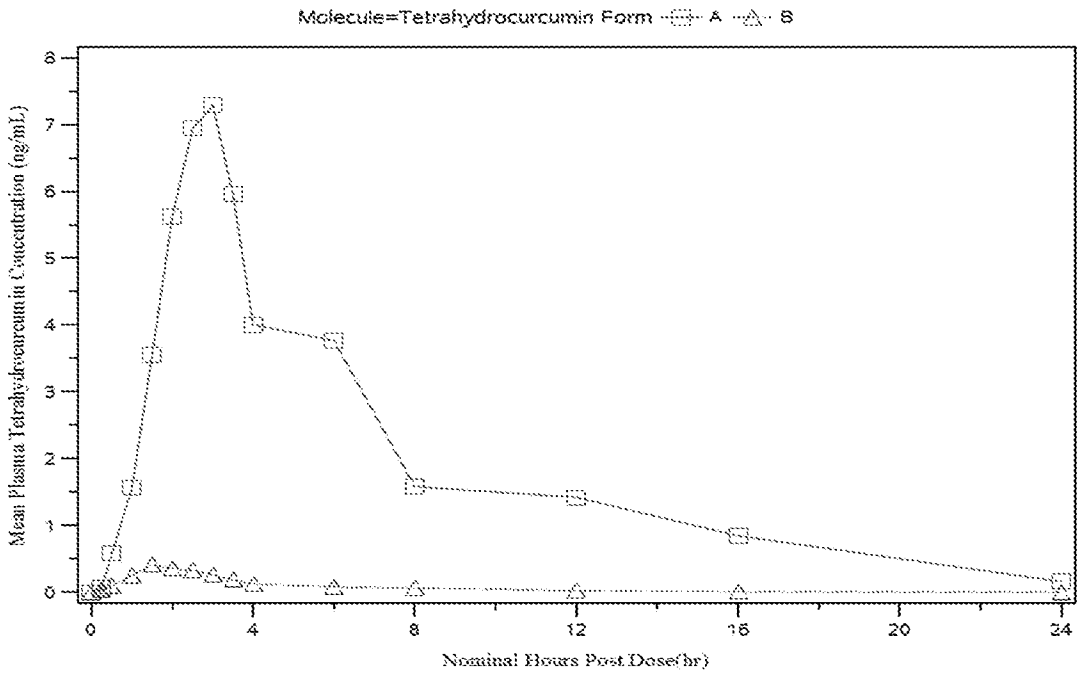
A = Test Product; B = Reference Product

Figure 4



A = Test Product; B = Reference Product

Figure 5



A = Test Product; B = Reference Product

Figure 6

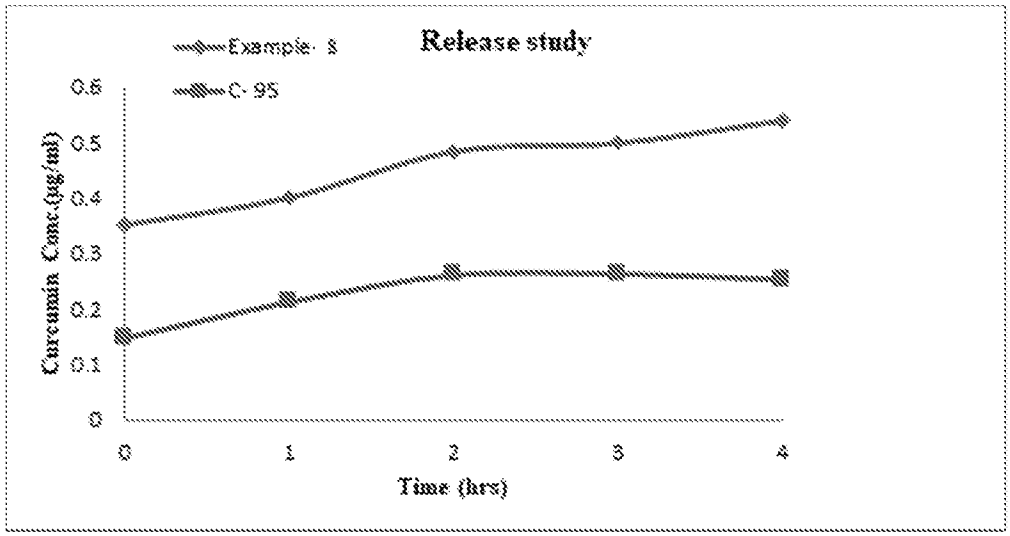


Figure 7

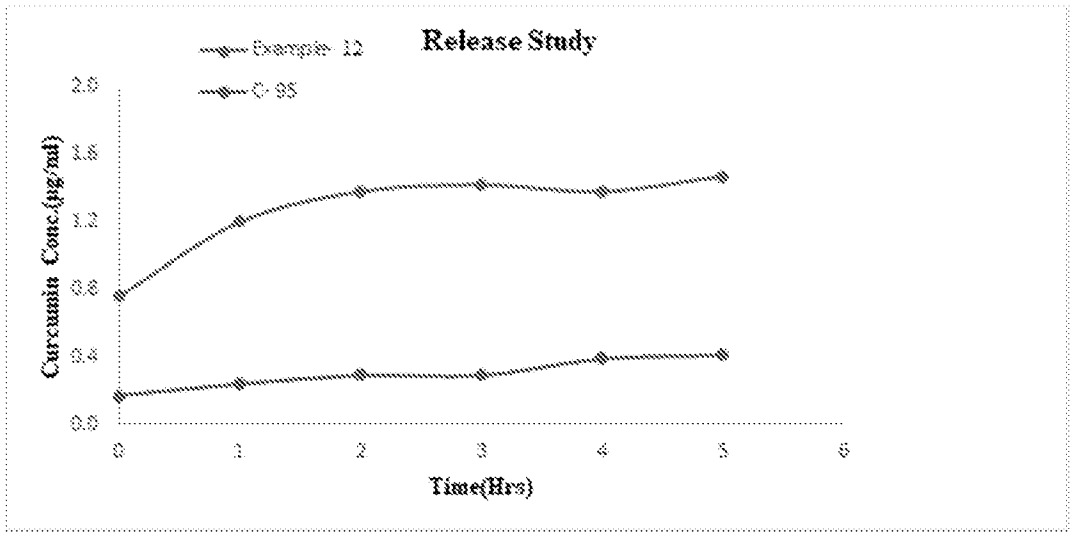


Figure 8

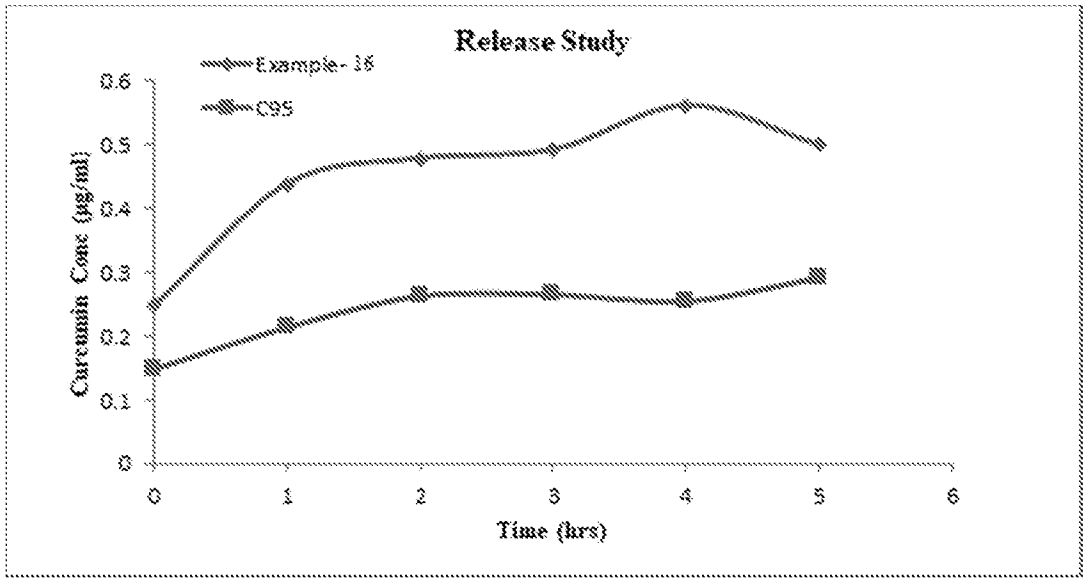


Figure 9

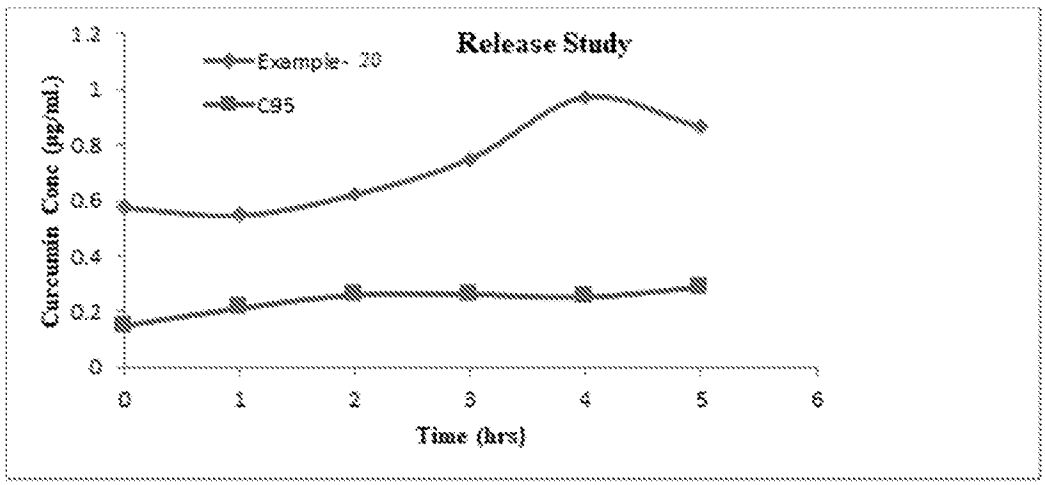


Figure 10

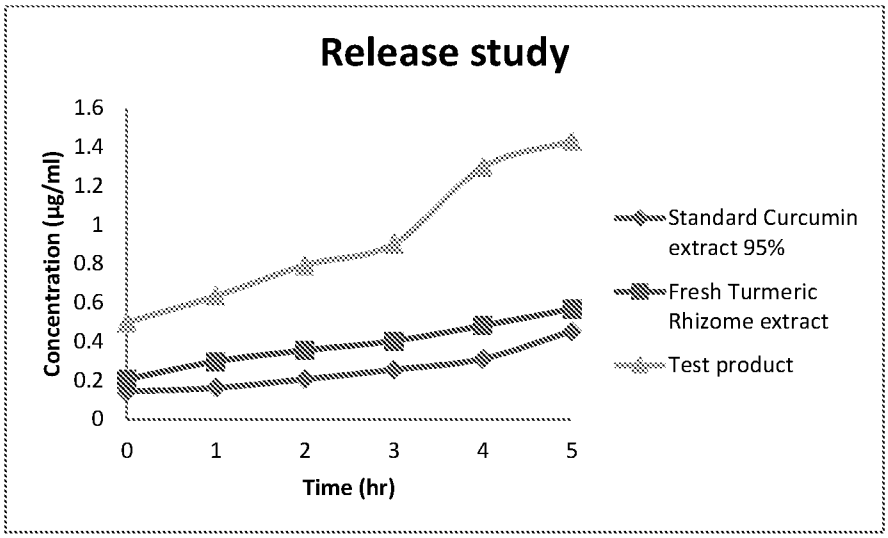


Figure 11

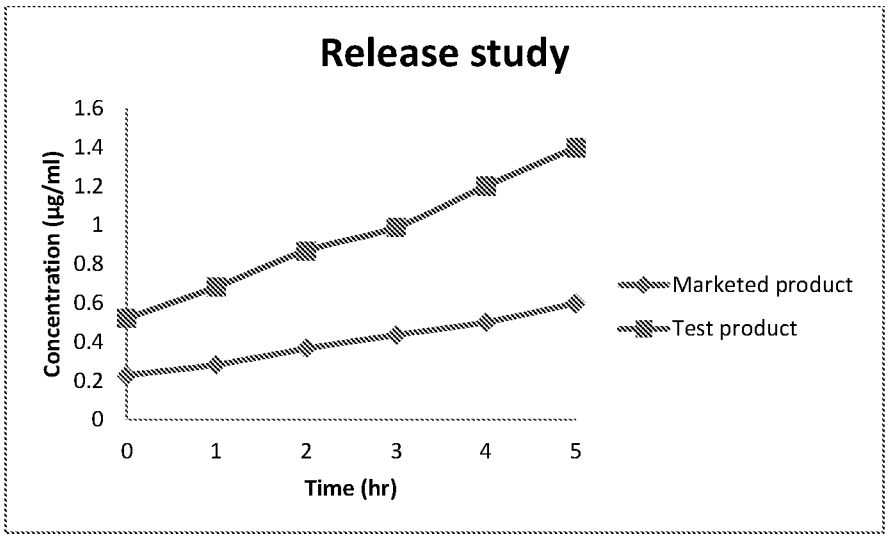


Figure 12

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IN2020/050519

A. CLASSIFICATION OF SUBJECT MATTER
A61K36/9066,A61K47/00,A61K31/00 Version=2020.01

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

Database: TotalPatent One, IPO Internal Database
Keywords: Curcuminoid, Curcumin, demethoxycurcumin, Bisdemethoxycurcumin.

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US20140010903A1; (AKAY FLAVOURS AND AROMATICS PVT LTD); Jan 09, 2014 (09-01-2014) *Refer whole document especially claims 1, 2 5 and 12, and paragraphs 55, 60 and 83*	1-5, 10
X	US8895087B2; (ARJUNA NATURAL EXTRACTS LTD); Nov 25, 2014 (25-11-2014) *Refer whole document especially lines 21-42 of column 4*	6-8

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
30-09-2020	30-09-2020

Name and mailing address of the ISA/	Authorized officer
Indian Patent Office Plot No.32, Sector 14,Dwarka,New Delhi-110075 Facsimile No.	Neha Prasad Telephone No. +91-1125300200

INTERNATIONAL SEARCH REPORT

International application No.
PCT/IN2020/050519

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☒ Claims Nos.: 9
because they relate to subject matter not required to be searched by this Authority, namely:
See continuation sheet

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☐ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/IN2020/050519

Citation	Pub.Date	Family	Pub.Date
US 8895087 B2	25-11-2014	ES 2709653 T3	17-04-2019
		WO 2006129323 A1	07-12-2006
		EP 1890546 B1	28-11-2018
		US 8153172 B2	10-04-2012
		PT 1890546 T	23-01-2019
		PL 1890546 T3	30-04-2019

International application No.
PCT/IN2020/050519

Continuation of Claims found unsearchable(Box II)

Claim 9 relates to a "method of treating inflammatory diseases, cognitive diseases, eye diseases, skin diseases and stress by administering therapeutically effective amount of bioavailable turmeric composition", which does not require an international search by the International Searching Authority in accordance with PCT Article 17(2)(a)(i) and [Rule 39.1(iv)].